

Tracing the Evolution of Popular Music Harmony:
A Computational Study Using Shannon Entropy

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Introduction

From the simple, infectious melodies of early rock and roll to today's genre-blurring chart-toppers, popular music has evolved in ways that reflect the shifting cultural landscape, shaping societal trends, fashion, and the evolution of other musical genres. Its distinctive sound has made it a subject of extensive study within music theory. However, analyses of this genre of music rely heavily on traditional frameworks, which is influenced on western music theory. Philip Tagg's *Analysing Popular Music: Theory, Method, and Practice* critiques the limitations of traditional western music theory in addressing the stylistic diversity of popular music.¹ While traditional frameworks provide insights, they fail to address the distinctive harmonic practices that define this music. These limitations highlight the need for approaches that better reflect the complexity and diversity of popular music.

Advancements in computational musicology offer promising tools for examining large-scale musical trends and patterns that may not be apparent through conventional analysis. In Trevor De Clercq and David Temperley's essay "A Corpus Analysis of Rock Music," they utilize computational methods to examine harmonic structures across transcribed musical datasets.² Their study identified chord progressions and harmonic patterns, providing statistical insights into chord progressions pattern, chord-to-chord relationships, and popular harmonic choices. Similarly, Lavengood applied topic modeling techniques to popular music analysis, exploring how timbre and production techniques relate to winter topics in video game music.³ These studies demonstrate how computational musicology can extend our understanding of musical harmony by revealing intricate patterns that traditional methods miss. Computational

¹ Philip Tagg, "Analysing Popular Music: Theory, Method, and Practice," *Popular Music* 2 (1982): 37-67.

² Trevor De Clercq and David Temperley, "A Corpus Analysis of Rock Music," *Popular Music* 30, no. 1 (2011): 47-70, <https://doi.org/10.1017/S026114301000067X>.

³ Megan Lavengood, "The Common Cold: Using Computational Musicology to Define the Winter Topic in Video Game Music," *Music Theory Online* 29, no. 1 (2023), <https://doi.org/10.30535/mt0.29.1.2>.

methods enable the processing of large and diverse musical datasets, allowing researchers to quantify chord-to-chord relationships, and identify chord progression patterns. However, while these corpus studies have expanded our understanding of harmony, these studies focus primarily on static chord-to-chord relationships or specific harmonic tags. Such studies neglect broader questions of harmonic complexity and how it evolves over time.

Information theory has been applied to music analysis in examining melodic and rhythmic structures; its application to harmonic analysis in popular music remains underexplored. Tuomas Eerola's study "Expectancy-Violation and Information-Theoretic Models of Melodic Complexity" employs information-theoretic models to understand how melodic complexity influences listener expectations, focusing on statistical properties of melodies rather than harmonic content.⁴ In "Evolution of the Informational Complexity of Contemporary Western Music," Manaris et al. analyze the informational complexity of Western music compositions over time using entropy, emphasizing mathematical modeling without musical interpretation.⁵ Additionally, Rosso et al.'s work "Complexity–Entropy Causality Plane: A Useful Approach for Distinguishing Songs" introduces a method for classifying songs based on complexity and entropy measures, offering a mathematical framework that prioritizes statistical analysis over the exploration of musical meaning or cultural context.⁶ While these studies advance the application of information theory in music analysis, they focus on quantitative methods and mathematical descriptions, without delving into musical interpretation or

⁴ Tuomas Eerola, "Expectancy-Violation and Information-Theoretic Models of Melodic Complexity," *Empirical Musicology Review* 11, no. 1 (2016): 2–17, <https://doi.org/10.18061/emr.v11i1.4836>.

⁵ Andreas Manaris, Petros R. Skordalakis, and Christodoulos Doulas, "Evolution of the Informational Complexity of Contemporary Western Music," in *Proceedings of the 5th International Workshop on Human-Friendly Human-Computer Interfaces* (2005): 33–40.

⁶ Haroldo V. Ribeiro, Luciano Zunino, Renio S. Mendes, and Ervin K. Lenzi, "Complexity–Entropy Causality Plane: A Useful Approach for Distinguishing Songs," *Physica A: Statistical Mechanics and Its Applications* 391, no. 7 (2009): 2421–28, <https://doi.org/10.1016/j.physa.2011.12.009>.

connecting findings to social and cultural changes. This emphasis on mathematical modeling limits their relevance to understanding the artistic and stylistic of music.

This study contributes to the limitation of computational musicology by adopting methods from information theory to broader musical changes. I explore the evolution of harmonic complexity in popular music from the 1950s to the 2000s using Shannon entropy to quantify the harmonic complexity⁷ of chords. Our research explores the following questions: 1) Has harmonic complexity in popular music increased over the decades? 2) Do musical forms exhibit significantly different levels of harmonic complexity? To answer these questions, I analyze a dataset of 890 popular songs spanning five decades, from the 1950s to the 2000s. By calculating the Shannon entropy of chord sequences within these songs, I aim to uncover patterns and trends in harmonic complexity that may have been overlooked by traditional analytical methods. From the results, I argue that harmonic complexity in popular music has evolved from the 1950s to the 2000s, increasing up to the 1980s and then experiencing a decline. These shifts align with major stylistic movements within the genre, suggesting that popular music's harmony aligns with its historical trends. This study contributes to the understanding of popular music harmony by applying information theory methods, connecting Shannon Entropy to musical analysis. By moving beyond traditional chord-to-chord analyses, it advances computational musicology through a broader examination of how harmonic complexity evolves over time, thereby deepening the understanding of harmonic structures in popular music genres. By analyzing harmonic complexity over five decades, the study provides a novel perspective on popular music harmony, enriching our understanding of the genre's musical development. In the

⁷ In this context, harmonic complexity is defined as the level of unpredictability and variation in a song's chord progression. While harmonic complexity can encompass various factors such as the use of extended chords, modulations, or harmonic density, focusing on entropy allows us to quantify one aspect of complexity that is particularly amenable to computational analysis.

following sections, I will describe our dataset and methodology in detail, present the results of our analysis, and discuss their implications within the broader context of popular music's evolution.

Data Set and Method

To investigate the evolution of harmonic complexity, we utilized the McGill Billboard Project dataset—a comprehensive collection of chord annotations and structural metadata stored in txt format for popular songs. The McGill Billboard Project is an open source dataset that includes annotations for 890 songs spanning from the 1950s to the 2000s, allowing for an analysis of harmonic trends over approximately five decades. Compiled by music researchers at McGill University, the dataset provides a ground truth for audio chord recognition and music analysis. The annotations encompass chord progressions, song structures, meter, key information, and other musical attributes, making it a valuable resource for computational musicology. This dataset is particularly well-suited for our study due to its extensive coverage of popular music across several decades and its detailed harmonic information.⁸

⁸ One limitation of using the McGill Billboard dataset is its focus on commercially successful songs from the Billboard Hot 100 charts, which introduces a selection bias toward mainstream popular music. As a result, our analysis may not fully capture the harmonic complexity present in less popular or underground subgenres that are not represented in the chart data.

Example 1: Example of the harmony annotation for “You’ve got a Friend”

```
# title: You've got a Friend
# artist: Roberta Flack and Donny Hathaway
# metre: 4/4
# tonic: Ab

0.000000000 silence
0.255419501 A, intro, | Ab:maj | Db:maj/5 | Ab:maj | G:hdim7 C:7 |, (synth)
14.013514739 B, verse, | F:min | C:7/5 C:7 | F:min C:7/5 | F:min/b3 C:7/5 F:min . |, (voice)
25.853922902 | Bb:min7 | Eb:7 | Ab:maj | Ab:maj |
37.546666666 C, pre-chorus, | G:hdim7 | C:7 | F:min C:7/5 | F:min/b3 C:7/5 F:min . |
49.184761904 | Bb:min7 | C:min7 | Eb:11 | Eb:maj |
60.961020408 D, chorus, | Ab:maj | Ab:7 | Db:maj | Db:maj |
72.560770975 | Ab:maj | Ab:maj | Eb:7 | Eb:7 |
84.028004535 | Ab:maj | Ab:maj7 | Db:maj | Gb:maj Db:maj . . |
95.393854875 | Db:maj7 C:min7 | Bb:min7 | Ab:maj |
103.635941043 E, bridge, | Gb:maj(9) | Db:maj | Ab:maj | Ab:maj |
114.834331065 | Db:maj | Gb:7 | F:min7 | Bb:min7 | Eb:11 | Eb:maj |
131.200000000 C, chorus, | Ab:maj | Ab:7 | Db:maj | Db:maj |
142.793265306 | Ab:maj | Ab:maj | Eb:7 | Eb:7 |
153.905238095 | Ab:maj | Ab:maj7 | Db:maj | Gb:maj Db:maj . . |
164.970136054 | Db:maj7 C:min7 | Bb:min7 | ->
170.329773242 F, outro, | Ab:maj | Db:maj/5 |
175.945396825 | Ab:maj | Db:maj/5 |
181.297392290 | Ab:maj | Db:maj/5 |
186.794897959 | Ab:maj | Db:maj/5 |
192.145192743 fadeout, | Ab:maj | Db:maj/5 |
197.540045351 | Ab:maj | Db:maj/5 |, (voice)
202.321473922 silence
206.540272108 end
```

For each text file, the process extracts relevant chord sequences while removing unwanted labels such as section markers ('A', 'B', etc.) and symbols like bar lines ('|'). The chord data is further organized by identifying different song sections (e.g., verse, chorus, bridge) based on predefined keywords, allowing for a clear distinction between song parts. Each chord sequence is then stored in the format of song_id, year of publication, section name, and chords. To quantify harmonic complexity in popular music, we employed Shannon entropy, an information-theoretic measure that assesses the unpredictability within a sequence of symbols—in this case, chords. Shannon entropy provides a mathematical framework for measuring the amount of information produced by a stochastic process, making it suitable for analyzing sequences where the probability of each element's occurrence varies. In the context of musical chord progressions, Shannon entropy allows us to capture the degree of uncertainty associated with the sequence of chords used in a song. A higher entropy value indicates greater variety in the chord progression, reflecting higher harmonic complexity. Conversely, a lower

entropy value suggests a more predictable and repetitive chord sequence, indicating lower harmonic complexity.

Example 2: Selections from the processed data by sections

song_id	year	section	chord_sequence
1069	1974	intro	C:maj F:maj C:maj F:maj
1069	1974	verse	C:maj C:maj C:maj G:m F:maj C:maj C:maj C:maj G:m F:maj
1069	1974	verse	C:maj C:maj C:maj G:m F:maj F:maj F:maj E:m D:m G:maj C:maj
1069	1974	verse	C:maj C:maj C:maj G:m F:maj C:maj C:maj C:maj G:m F:maj
1069	1974	verse	C:maj C:maj C:maj G:m F:maj F:maj F:maj E:m D:m G:maj C:maj
1069	1974	chorus	C:maj F:maj C:maj C:maj D:m F:maj G:maj C:maj F:maj C:maj C:maj
1069	1974	chorus	C:maj F:maj C:maj C:maj D:m F:maj G:maj C:maj F:maj C:maj C:maj
1069	1974	verse	C:maj C:maj C:maj G:m F:maj C:maj C:maj C:maj G:m F:maj
1069	1974	verse	C:maj C:maj C:maj G:m F:maj F:maj F:maj E:m D:m G:maj C:maj
1069	1974	chorus	C:maj F:maj C:maj C:maj D:m F:maj G:maj C:maj F:maj C:maj C:maj
1069	1974	chorus	C:maj F:maj C:maj C:maj D:m F:maj G:maj C:maj F:maj C:maj C:maj
1069	1974	bridge	C:maj
1069	1974	bridge	F:maj C:maj F:maj C:maj F:maj

Although this algorithm is able to extract harmonic diversity in order to quantify the complexity of the harmony, this method has the following limitations. Complex chords were simplified to their basic triadic forms to maintain consistency across the dataset. While this standardization was necessary for computational analysis, it may slightly underestimate harmonic complexity by omitting chord alterations and voice leading. While Shannon entropy considers the probability distribution of chords, it does not account for the sequential order of chords. To account for differences in song lengths and section durations, entropy values were normalized where appropriate. This ensured that comparisons between songs and sections were valid and not biased by the amount of data in each case.

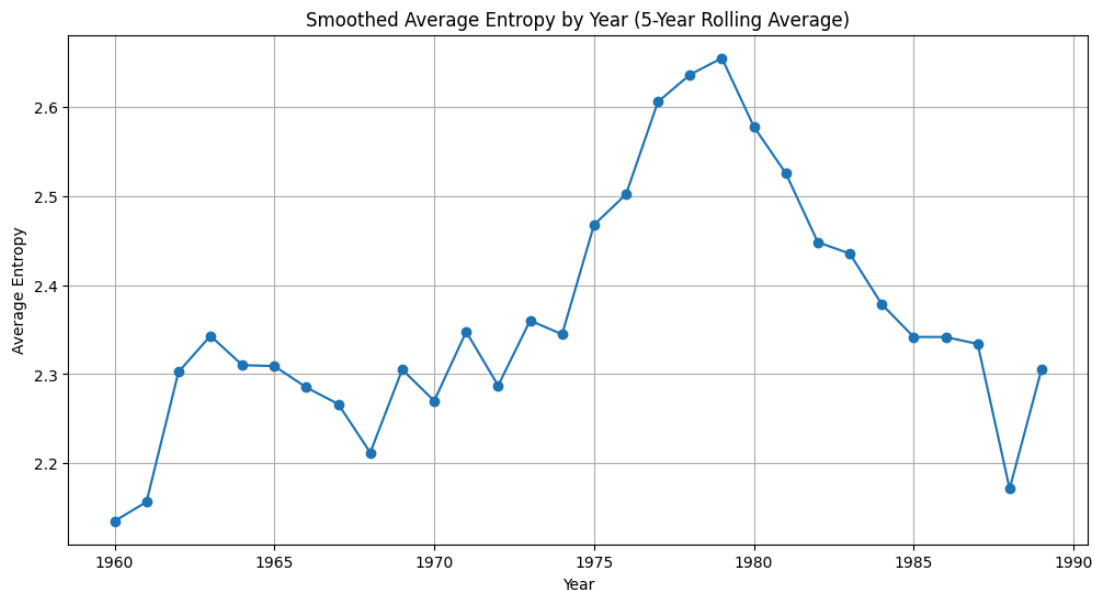
This analysis examines trends in harmonic complexity across time using two main visualizations: a 5-year rolling average line graph (Example 3) and an entropy distribution box plot by year (Example 4). Together, these graphs reveal how popular music's harmonic structure has responded to various stylistic shifts over the decades. Through these analyses, we observe

not only broad trends in complexity but also the diversity within each era, helping us understand how popular music's harmonic language reflects its artistic movements.

Complexity analyzed by Year

The 5-year rolling average line graph of harmonic complexity reveals significant long-term trends from the 1950s to the 2000s, smoothing out yearly fluctuations to highlight broader changes in popular music. The graph shows a steady increase in complexity before the 1980s, followed by a decline during that decade and ongoing fluctuations in the years that follow. The steady rise in harmonic complexity up to the late 1970s suggests a period of increased experimentation. Movements like R&B, psychedelic rock, and jazz fusion encouraged artists to push the boundaries of traditional structures. Bands such as Pink Floyd and The Beatles incorporated complex chord progressions and lengthy compositions, creating a rich harmonic landscape. This trend reflects popular music's spirit of innovation during the 1960s and 1970s, as artists drew from diverse influences to shape a new sound.

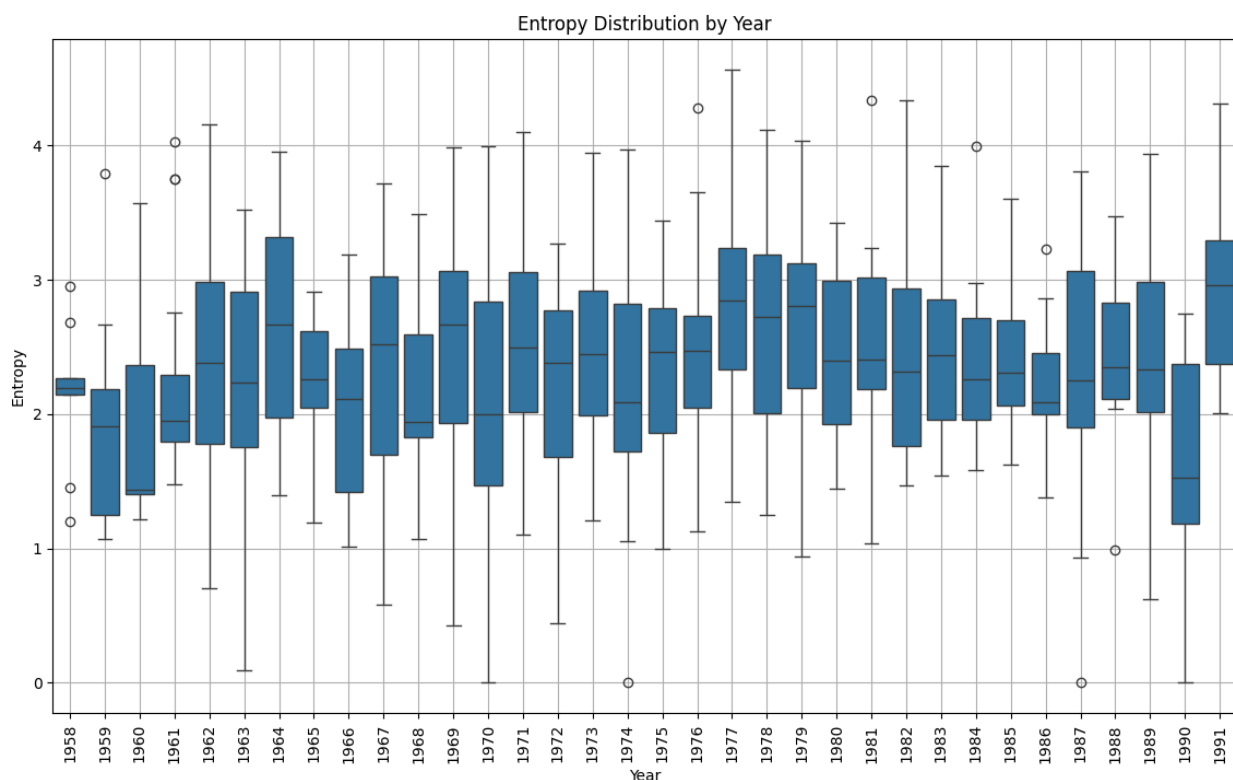
Example 3: Average Entropy by Year



The marked decline in harmonic complexity in the 1980s corresponds with the rise of punk rock, hit-pop, and early electronic music. These genres emphasized simpler, more direct structures, often in response to the perceived excess of the 1970s. Bands like The Clash, Talking Heads, and Journey popularized this back-to-basics approach, with harmonic structures that favored accessibility and energy over complexity. The simplicity of this decade reflects the cultural shift toward more radio-friendly and commercially viable music, prioritizing catchy, anthemic sounds that resonated with a wide audience. Complexity begins to fluctuate in the 1990s, showing periods of higher and lower harmonic entropy as popular music diversifies. The rise of grunge and alternative rock brought an emphasis on raw, emotional expression, with bands like Nirvana and Coldplay embracing both straightforward and innovative structures. The continued fluctuations in complexity during the 2000s reflect the genre's increased eclecticism, with indie rock and genre-blending allowing for both minimalist and intricate harmonic approaches. This trend suggests that popular music's harmonic complexity remains fluid, adapting to each era's cultural landscape and technological advancements.

The entropy distribution by year box plot (figure 4) provides additional insights by highlighting the variability and consistency of harmonic complexity within each year. Unlike the average-based line graph, the box plot reveals how diverse or uniform harmonic structures were within each era, showing how movements affected not only the average complexity but also the range of musical expression.

Example 4: Entropy Distribution by Year



In the 1960s and 1970s, the box plot shows a broad range of entropy values, with high variability in harmonic complexity. This wide distribution reflects the diverse approaches to composition during this experimental era, where music embraced eclectic influences from blues, jazz, and classical music. Progressive and psychedelic rock movements encouraged both intricate and straightforward harmonies, resulting in a variety of harmonic structures within a single

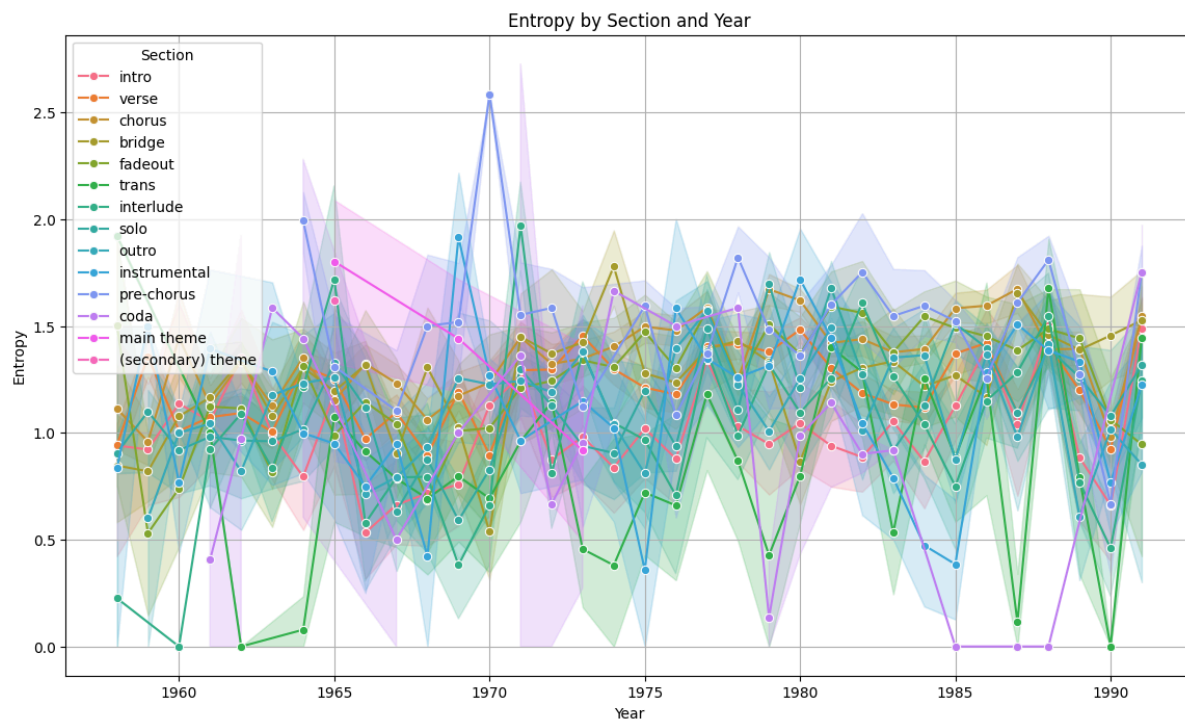
decade. The 1980s show a narrowing in the entropy, with fewer outliers and a more clustered range of values. This uniformity aligns with the popularity of genres that emphasized simplicity and accessibility. The reduced variability suggests a standardization in popular music, as bands adopted similar harmonic structures that resonated with the mainstream audience. With artists like The Police and U2, the focus was on catchy hooks and straightforward harmonic progressions, catering to radio-friendly formats and large-scale performances. This convergence reflects the 1980s shift toward a more commercially driven music landscape. In the 1990s, the box plot reveals a renewed increase in harmonic variability, with a wider range of entropy values. The broader distribution of entropy in this era indicates a blend of simpler and more complex harmonic approaches, reflecting the 1990s ethos of authenticity and emotional expression. This era's diversity in harmonic complexity underscores popular music's ability to incorporate various influences while challenging the mainstream trends of previous decades.

In conclusion, the analysis of harmonic complexity through both the 5-year rolling average and the entropy distribution by year reveals how popular music's structure reflects its historic movements. The pre-1980s increase in complexity and wide variability suggest a period of innovation and diversity, as artists explored new harmonic ideas. The 1980s decline in complexity and narrowing of distribution show a shift toward simplicity and uniformity. The renewed variability in the 1990s and the diverse range in recent decades highlight popular music's adaptability and its response to evolving cultural landscapes. These trends indicate that harmonic complexity in popular music is not only a technical trait but a reflection of the genre's engagement with musical changes.

Complexity Analysis by Sections

In addition to exploring overall trends in harmonic complexity, analyzing entropy by song section (intro, verse, chorus, etc.) provides insights into how specific parts of a song contribute to popular music's evolving harmonic language. Each section serves a unique role in a song's structure: intros often set the tonal mood, verses provide narrative, and choruses are typically memorable and hook-driven. Examining entropy by section reveals how harmonic complexity varies within these parts, shedding light on stylistic shifts within popular music's compositional approach.

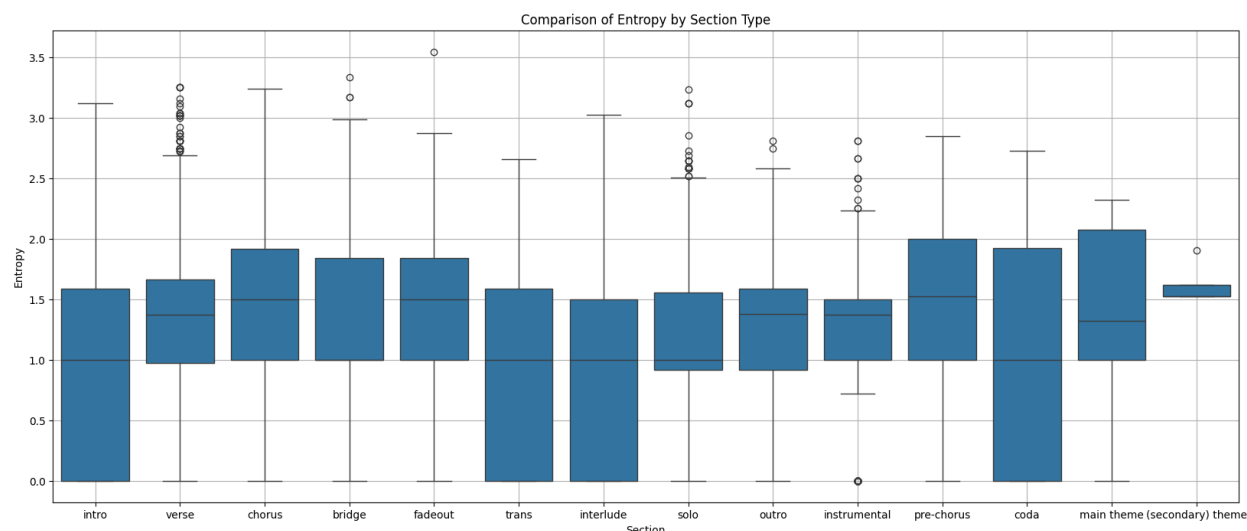
Figure 5: Entropy by Section and year



Using the Entropy line graph that shows average entropy by section across different years, we can track how harmonic complexity in intros, verses, choruses, and other sections has evolved over time. Intros often show lower harmonic complexity than other sections, likely due

to their function of establishing a song's initial tone and drawing the listener in with a familiar or repetitive structure. In the pre-1980s era, music used complex intros to set an exploratory tone, while later decades, such as the 1980s, simplified intros as part of a move toward radio-friendly formats. Intros became more straightforward, reflecting the era's emphasis on immediacy and accessibility. In recent years, intros may vary more widely in complexity, with some artists using minimalism to captivate listeners and others incorporating intricate arrangements for unique hooks.

Verses typically exhibit moderate complexity, as they carry the song's narrative and often employ more harmonic variety than intros. The graph may show slight rises and dips in verse complexity that mirror general trends in popular music's harmonic evolution. In the 1960s and 1970s, verses were often elaborate to support storytelling. However, during the 1980s, verse complexity dropped as people favored straightforward, relatable storytelling. In the 1990s, complexity in verses varied widely, with alternative rock and grunge bringing rawness, while some bands like Radiohead experimented with complex harmonic structures. Choruses frequently show lower entropy than verses, as they are often the most repetitive, memorable part of a song. However, some years exhibit an increase in chorus complexity, possibly reflecting shifts in how bands approach this central section. Choruses have historically been hook-driven, particularly in the 1980s when catchy, anthemic choruses became the norm. The 1990s and 2000s introduced more experimentation in choruses, as alternative rock and indie bands varied harmonic structures to stand out. Bridges and transitions, when included, often exhibit higher complexity than other sections, adding harmonic contrast within a song. The graph might reveal peaks in bridge entropy that vary over time

Figure 6: Comparison of Entropy by Section Type

The entropy distribution by section, shown in box plots, provides insights into the range of harmonic complexity within each section. This distribution reveals which sections have more variability and how different song parts contribute to the overall harmonic landscape of popular music. The distribution plot for intros may show a narrower range in entropy, especially in later decades, as intros became more standardized. Intros with low variability suggest a preference for simple, mood-setting openings that ease the listener into the song. Verses tend to have a moderate range of entropy values, reflecting their role in carrying the song's story while allowing for some harmonic diversity. Choruses generally show lower entropy variability, as they often emphasize repetition and memorability. Consistently lower variability in choruses, from the 1980s onward, underscores popular music's emphasis on catchy, singable hooks during this period. The slight increase in chorus variability in the 1990s and 2000s corresponds with its willingness to diverge from standard formats, allowing for more harmonic complexity in choruses while still delivering a central hook. Bridges often have the widest range of entropy values, reflecting their purpose as contrasting, transitional elements within a song. The high

variability in bridge entropy suggests that artists use this section to introduce complexity, providing a break from the main structure.

In analyzing entropy by song section, we observe that each section has a unique role in shaping a song's harmonic complexity. The trends in section-specific entropy reveal how popular music's compositional approach has evolved. This section-based analysis highlights how popular music's harmonic complexity is not just a reflection of overall trends but also a dynamic characteristic influenced by each song's purpose within the genre's changing landscape. Together, these findings underscore the adaptability and richness of popular music, showing how it balances simplicity and complexity across different song sections in response to stylistic shifts.

Conclusion

This study has explored the evolution of harmonic complexity in popular music from the 1950s to the 2020s by applying Shannon entropy as a quantitative measure. Our analysis revealed that harmonic complexity increased significantly up to the 1980s, followed by a decline in subsequent decades. These fluctuations correspond closely with stylistic shifts within the genre. Movements such as the experimentalism of the 1970s introduced greater harmonic complexity, while the emergence of more genres emphasized simplicity and raw emotion, leading to a decrease in harmonic complexity. Each phase brings its own ethos, influencing harmonic structures to align with broader cultural and musical trends. By moving beyond traditional chord-to-chord analyses, this study contributes to computational musicology by offering a more nuanced understanding of harmonic complexity in popular music. Applying information theory through Shannon entropy directly connects quantitative metrics with musical interpretation, providing a novel way to understand how popular music harmony has evolved

over time. This approach breaks new ground in the study of popular music's harmonic structures, challenging the perception of popular music harmony as simplistic or static and highlighting its dynamic progression and sophistication.

Future research could expand upon these findings by incorporating additional musical elements into the analysis. Exploring complexity measures for basslines, rhythmic patterns, and song forms could provide a more comprehensive understanding of musical complexity in popular music. Additionally, investigating whether harmonic complexity correlates with factors such as popularity, critical acclaim, or cultural impact might offer valuable insights into the relationship between musical complexity and a song's reception. Such multidimensional analyses would deepen our understanding of popular music's evolution and its role within the broader landscape of popular music.

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